

Magnetism and Matter

Chapter 5 · All Questions and Solutions



Part 1 · Examples

Q

1

Cutting and testing a bar magnet



(a) What happens if a bar magnet is cut into two pieces: (i) transverse to its length, (ii) along its length? (b) A magnetised needle in a uniform magnetic field experiences a torque but no net force. An iron nail near a bar magnet, however, experiences a force of attraction in addition to a torque. Why? (c) Must every magnetic configuration have a north pole and a south pole? What about the field due to a toroid? (d) Two identical looking iron bars A and B are given, one of which is definitely known to be magnetised. (We do not know which one.) How would one ascertain whether or not both are magnetised? If only one is magnetised, how does one ascertain which one? [Use nothing else but the bars A and B.]

Part (a)

What happens if a bar magnet is cut into two pieces: (i) transverse to its length, (ii) along its length?

Conclusion

In either case, one gets two magnets, each with a north and south pole.

Part (b)

A magnetised needle in a uniform magnetic field experiences a torque but no net force. An iron nail near a bar magnet, however, experiences a force of attraction in addition to a torque. Why?

Conclusion

No force if the field is uniform.

The iron nail experiences a nonuniform field due to the bar magnet.

There is induced magnetic moment in the nail, therefore, it experiences both force and torque.

The net force is attractive because the induced south pole (say) in the nail is closer to the north pole of magnet than induced north pole.

Part (c)

Must every magnetic configuration have a north pole and a south pole? What about the field due to a toroid?

Conclusion

Not necessarily.

True only if the source of the field has a net non-zero magnetic moment.

This is not so for a toroid or even for a straight infinite conductor.

Part (d)

Two identical looking iron bars A and B are given, one of which is definitely known to be magnetised. (We do not know which one.) How would one ascertain whether or not both are magnetised? If only one is magnetised, how does one ascertain which one? [Use nothing else but the bars A and B.]

Conclusion

Bring different ends of the two bars close together.

If you get a repulsive force in some orientation, both bars are magnetised.

If the force is always attractive, one of them is not magnetised.

In a bar magnet, the field is strongest at the two ends (poles) and weakest at the centre.

This lets us find which bar, A or B, is the magnet: pick up one bar (say A) and bring one of its ends near an end of the other bar (B), then near the middle of B.

If A feels no force at the middle of B, then B is the magnet.

If the force does not change from the end to the middle of B, then A is the magnet.

Q

2

Equilibrium of a needle in another needle's field



Figure 5.4 shows a small magnetised needle P placed at a point O. The arrow shows the direction of its magnetic moment. The other arrows show different positions (and orientations of the magnetic moment) of another identical magnetised needle Q. (a) In which configuration the system is not in equilibrium? (b) In which configuration is the system in (i) stable, and (ii) unstable equilibrium? (c) Which configuration corresponds to the lowest potential energy among all the configurations shown?

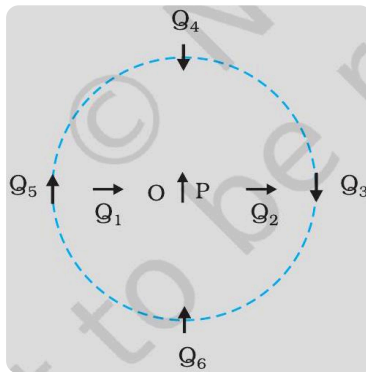


FIGURE 5.4

Part (a)

In which configuration the system is not in equilibrium?

Given

Potential energy of the configuration arises due to the potential energy of one dipole (say, Q) in the magnetic field due to other (P).

Use the result that the field due to P is given by the expression [Eqs. (5.4) and (5.5)]:

Key formula

$$\mathbf{B}_P = -\frac{\mu_0}{4\pi} \frac{\mathbf{m}_P}{r^3} \quad (\text{on the normal bisector})$$

$$\mathbf{B}_P = \frac{\mu_0}{4\pi} \frac{2\mathbf{m}_P}{r^3} \quad (\text{on the axis})$$

where $\mathbf{m}_P$ is the magnetic moment of the dipole P.

Conclusion

Equilibrium is stable when $\mathbf{m}_Q$ is parallel to $\mathbf{B}_P$, and unstable when it is anti-parallel to $\mathbf{B}_P$.

PQ₁ and PQ₂

Part (b)

In which configuration is the system in (i) stable, and (ii) unstable equilibrium?

Conclusion(i) PG_3 , PO_6 (stable); (ii) PQ_5 , PQ_4 (unstable)

Part (c)

Which configuration corresponds to the lowest potential energy among all the configurations shown?

Conclusion PO_6 **Q****3**

Spotting
wrong field-
line
diagrams

(())

Many of the diagrams given in Fig. 5.6 show magnetic field lines (thick lines in the figure) wrongly. Point out what is wrong with them. Some of them may describe electrostatic field lines correctly. Point out which ones.

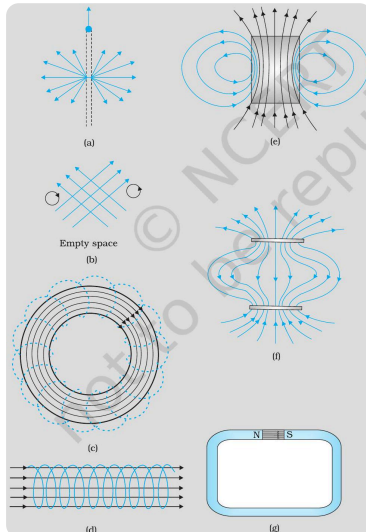


FIGURE 5.6

Conclusion

(a) Wrong.

Magnetic field lines can never emanate from a point, as shown in figure.

Over any closed surface, the net flux of $\mathbf{B}$ must always be zero, i.e., pictorially as many field lines should seem to enter the surface as the number of lines leaving it. The field lines shown, in fact, represent electric field of a long positively charged wire.

The correct magnetic field lines are circling the straight conductor, as described in Chapter 4.

(b) Wrong.

Magnetic field lines (like electric field lines) can never cross each other, because otherwise the direction of field at the point of intersection is ambiguous.

There is further error in the figure.

Magnetostatic field lines can never form closed loops around empty space.

A closed loop of static magnetic field line must enclose a region across which a current is passing.

By contrast, electrostatic field lines can never form closed loops, neither in empty space, nor when the loop encloses charges.

(c) Right.

Magnetic lines are completely confined within a toroid.

Nothing wrong here in field lines forming closed loops, since each loop encloses a region across which a current passes.

Note, for clarity of figure, only a few field lines within the toroid have been shown.

Actually, the entire region enclosed by the windings contains magnetic field.

(d) Wrong.

Field lines due to a solenoid at its ends and outside cannot be so completely straight and confined; such a thing violates Ampere's law.

The lines should curve out at both ends, and meet eventually to form closed loops.

(e) Right.

These are field lines outside and inside a bar magnet.

Note carefully the direction of field lines inside.

Not all field lines emanate out of a north pole (or converge into a south pole).

Around both the N-pole, and the S-pole, the net flux of the field is zero.

(f) Wrong.

These field lines cannot possibly represent a magnetic field.

Look at the upper region.

All the field lines seem to emanate out of the shaded plate.

The net flux through a surface surrounding the shaded plate is not zero.

This is impossible for a magnetic field.

The given field lines, in fact, show the electrostatic field lines around a positively charged upper plate and a negatively charged lower plate.

The difference between Fig. [5.6(e) and (f)] should be carefully grasped.

(g) Wrong.

Magnetic field lines between two pole pieces cannot be precisely straight at the ends.

Some fringing of lines is inevitable.

Otherwise, Ampere's law is violated.

This is also true for electric field lines.

(a),(b),(d),(f),(g) wrong; (c),(e) right //

Q

4

Conceptual
questions on
field lines
and
monopoles



Can a system have magnetic moments even though its net charge is zero?

Part (a)

Magnetic field lines show the direction (at every point) along which a small magnetised needle aligns (at the point). Do the magnetic field lines also represent the lines of force on a moving charged particle at every point?

Conclusion

No.

The magnetic force is always normal to $\mathbf{B}$ (remember magnetic force $= q \mathbf{v} \times \mathbf{B}$).

It is misleading to call magnetic field lines as lines of force.

Part (b)

If magnetic monopoles existed, how would the Gauss's law of magnetism be modified?

Conclusion

Gauss's law of magnetism states that the flux of $\mathbf{B}$ through any closed surface is always zero $\oint_S \mathbf{B} \cdot \Delta \mathbf{s} = 0$.

If monopoles existed, the right hand side would be equal to the monopole (magnetic charge) q_m enclosed by S.

[Analogous to Gauss's law of electrostatics, $\oint_S \mathbf{B} \cdot \Delta \mathbf{s} = \mu_0 q_m$ where q_m is the (monopole) magnetic charge enclosed by S.]

Part (c)

Does a bar magnet exert a torque on itself due to its own field? Does one element of a current-carrying wire exert a force on another element of the same wire?

Conclusion

No.

There is no force or torque on an element due to the field produced by that element itself.

But there is a force (or torque) on an element of the same wire.

(For the special case of a straight wire, this force is zero.)

Part (d)

Magnetic field arises due to charges in motion. Can a system have magnetic moments even though its net charge is zero?

Conclusion

Yes.

The average of the charge in the system may be zero.

Yet, the mean of the magnetic moments due to various current loops may not be zero.

We will come across such examples in connection with paramagnetic material where atoms have net dipole moment though their net charge is zero.

Q

5

Field, magnetisation and current in a solenoid with a core



A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate (a) H, (b) M, (c) B and (d) the magnetising current I_m .

Given

(a) The field H is dependent of the material of the core, and is $H = n I$
 $I = 1000 \times 2.0 = 2 \times 10^3 \text{ A / m.}$

Key formula

(b) The magnetic field B is given by

$$\begin{aligned} B &= \mu_r \mu_0 H \\ &= 400 \times 4\pi \times 10^{-7} (\text{N / A}^2) \times 2 \times 10^3 \frac{\text{A}}{\text{m}} \\ &= 1.0 \text{ T} \end{aligned}$$

Conclusion

(c) Magnetisation is given by

$$\begin{aligned} M &= \frac{(B - \mu_0 H)}{\mu_0} \\ &= \frac{(\mu_r \mu_0 H - \mu_0 H)}{\mu_0} = (\mu_r - 1) H = 399 \times H \\ &\cong 8 \times 10^5 \text{ A / m} \end{aligned}$$

(d) The magnetising current I_m is the additional current that needs to be passed through the windings of the solenoid in the absence of the core which would give a B

value as in the presence of the core.

Thus $B = \mu_r n(I + I_M)$.

Using $I = 2 \text{ A}$, $B = 1 \text{ T}$, we get $I_M = 794 \text{ A}$.

$$\mathbf{H = 2 \times 10^3 \text{ A/m; } B = 1.0 \text{ T; } M \cong 8 \times 10^5 \text{ A/m; } I_M = 794 \text{ A}} //$$

Q

1

Magnetic moment of a bar magnet from torque



A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude equal to $4.5 \times 10^{-2} \text{ J}$. What is the magnitude of magnetic moment of the magnet?

Given

Magnetic field strength, $B = 0.25 \text{ T}$

Torque on the bar magnet, $T = 4.5 \times 10^{-2} \text{ J}$

Angle between the bar magnet and the external magnetic field, $\theta = 30^\circ$

Key formula

Torque is related to magnetic moment (M) as:

$$T = M B \sin \theta$$

$$\therefore M = \frac{T}{B \sin \theta}$$

$$= \frac{4.5 \times 10^{-2}}{0.25 \times \sin 30^\circ} = 0.36 \text{ J T}^{-1}$$

Conclusion

Hence, the magnetic moment of the magnet is 0.36 J T^{-1} .

0.36 J T⁻¹ //

Q

2

Stable and unstable equilibrium of a bar magnet

(())

A short bar magnet of magnetic moment $m = 0.32 \text{ JT}^{-1}$ is placed in a uniform magnetic field of 0.15 T . If the bar is free to rotate in the plane of the field, which orientation would correspond to its (a) stable, and (b) unstable equilibrium? What is the potential energy of the magnet in each case?

Part (a)

Stable equilibrium orientation - what orientation, and what is the potential energy of the magnet in this case?

Part (b)

Unstable equilibrium orientation - what orientation, and what is the potential energy of the magnet in this case?

Conclusion

Moment of the bar magnet, $M = 0.32 \text{ J T}^{-1}$

External magnetic field, $B = 0.15 \text{ T}$

(a) The bar magnet is aligned along the magnetic field.

This system is considered as being in stable equilibrium.

Hence, the angle θ , between the bar magnet and the magnetic field is 0° .

Potential energy of the system $= -M B \cos \theta$

$$= -0.32 \times 0.15 \cos 0^\circ$$

$$= -4.8 \times 10^{-2} \text{ J}$$

(b) The bar magnet is oriented 180° to the magnetic field.

Hence, it is in unstable equilibrium.

$$\theta = 180^\circ$$

Potential energy $= -M B \cos \theta$

$$= -0.32 \times 0.15 \cos 180^\circ$$

$$= 4.8 \times 10^{-2} \text{ J}$$

stable: $\theta = 0^\circ$, $U = -4.8 \times 10^{-2} \text{ J}$; unstable: $\theta = 180^\circ$, $U = 4.8 \times 10^{-2} \text{ J}$ ///

Q

3

Magnetic moment of a current-carrying solenoid



A closely wound solenoid of 800 turns and area of cross section $2.5 \times 10^{-4} \text{ m}^2$ carries a current of 3.0 A. Explain the sense in which the solenoid acts like a bar magnet. What is its associated magnetic moment?

Given

Number of turns in the solenoid, $n = 800$

Area of cross-section, $A = 2.5 \times 10^{-4} \text{ m}^2$

Current in the solenoid, $I = 3.0 \text{ A}$

A current-carrying solenoid behaves as a bar magnet because a magnetic field develops along its axis, i.e., along its length.

Key formula

The magnetic moment associated with the given current-carrying solenoid is calculated as:

$$M = n I A$$

$$= 800 \times 3 \times 2.5 \times 10^{-4}$$

$$= 0.6 \text{ J T}^{-1}$$

0.6 J T^{-1} ///

Q

4

Torque on a solenoid turning about a vertical axis



If the solenoid in Exercise 5.5 is free to turn about the vertical direction and a uniform horizontal magnetic field of 0.25 T is applied, what is the magnitude of torque on the solenoid when its axis makes an angle of 30° with the direction of applied field?

Given

Magnetic field strength, $B = 0.25 \text{ T}$

Magnetic moment, $M = 0.6 \text{ T}^{-1}$

The angle θ , between the axis of the solenoid and the direction of the applied field is 30° .

Key formula

Therefore, the torque acting on the solenoid is given as:

$$\tau = M B \sin \theta$$

$$= 0.6 \times 0.25 \sin 30^\circ$$

$$= 7.5 \times 10^{-2} \text{ J}$$

$7.5 \times 10^{-2} \text{ J}$ ///

Q

5

Work and
torque
turning a bar
magnet in a
field



A bar magnet of magnetic moment 1.5 J T^{-1} lies aligned with the direction of a uniform magnetic field of 0.22 T . (a) What is the amount of work required by an external torque to turn the magnet so as to align its magnetic moment: (i) normal to the field direction, (ii) opposite to the field direction? (b) What is the torque on the magnet in cases (i) and (ii)?

Part (a)

What is the amount of work required by an external torque to turn the magnet so as to align its magnetic moment: (i) normal to the field direction, (ii) opposite to the field direction?

Part (b)

What is the torque on the magnet in cases (i) and (ii)?

Conclusion

(a) Magnetic moment, $M = 1.5 \text{ J T}^{-1}$

Magnetic field strength, $B = 0.22 \text{ T}$

(i) Initial angle between the axis and the magnetic field, $\theta_1 = 0^\circ$

Final angle between the axis and the magnetic field, $\theta_2 = 90^\circ$

The work required to make the magnetic moment normal to the direction of magnetic field is given as:

$$\begin{aligned} W &= -M B (\cos \theta_2 - \cos \theta_1) \\ &= -1.5 \times 0.22 (\cos 90^\circ - \cos 0^\circ) \\ &= -0.33(0-1) \\ &= 0.33 \text{ J} \end{aligned}$$

(ii) Initial angle between the axis and the magnetic field, $\theta_1 = 0^\circ$

Final angle between the axis and the magnetic field, $\theta_2 = 180^\circ$

The work required to make the magnetic moment opposite to the direction of magnetic field is given as:

$$\begin{aligned} W &= -M B (\cos \theta_2 - \cos \theta_1) \\ &= -1.5 \times 0.22 (\cos 180^\circ - \cos 0^\circ) \\ &= -0.33(-1-1) \\ &= 0.66 \text{ J} \end{aligned}$$

(b) For case (i): $\theta = \theta_2 = 90^\circ$

$$\begin{aligned} \therefore \text{Torque, } \tau &= M B \sin \theta \\ &= 1.5 \times 0.22 \sin 90^\circ \\ &= 0.33 \text{ J} \end{aligned}$$

For case (ii): $\theta = \theta_2 = 180^\circ$

$$\begin{aligned} \therefore \text{Torque, } \tau &= M B \sin \theta \\ &= M B \sin 180^\circ = 0 \text{ J} \end{aligned}$$

(i) $W = 0.33 \text{ J}$, (ii) $W = 0.66 \text{ J}$; torque (i) $= 0.33 \text{ J}$, torque (ii) $= 0 \text{ J}$



Q

6

Magnetic moment, force and torque on a suspended solenoid

()

A closely wound solenoid of 2000 turns and area of cross-section $1.6 \times 10^{-4} \text{ m}^2$, carrying a current of 4.0 A, is suspended through its centre allowing it to turn in a horizontal plane.

Part (a)

What is the magnetic moment associated with the solenoid?

Part (b)

What is the force and torque on the solenoid if a uniform horizontal magnetic field of $7.5 \times 10^{-2} \text{ T}$ is set up at an angle of 30° with the axis of the solenoid?

Conclusion

Number of turns on the solenoid, $n = 2000$

Area of cross-section of the solenoid, $A = 1.6 \times 10^{-4} \text{ m}^2$

Current in the solenoid, $I = 4 \text{ A}$

(a) The magnetic moment along the axis of the solenoid is calculated as:

$$\begin{aligned} M &= n A I \\ &= 2000 \times 1.6 \times 10^{-4} \times 4 \\ &= 1.28 \text{ Am}^2 \end{aligned}$$

(b) Magnetic field, $B = 7.5 \times 10^{-2} \text{ T}$

Angle between the magnetic field and the axis of the solenoid, $\theta = 30^\circ$

Torque, $\tau = M B \sin \theta$

$$\begin{aligned} &= 1.28 \times 7.5 \times 10^{-2} \sin 30^\circ \\ &= 4.8 \times 10^{-2} \text{ Nm} \end{aligned}$$

Since the magnetic field is uniform, the force on the solenoid is zero.

The torque on the solenoid is $4.8 \times 10^{-2} \text{ Nm}$.

$$M = 1.28 \text{ A m}^2; \text{ force zero; torque } 4.8 \times 10^{-2} \text{ N m} //$$

Q

7

Axial and equatorial field of a bar magnet



A short bar magnet has a magnetic moment of 0.48 J T^{-1} . Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on (a) the axis, (b) the equatorial lines (normal bisector) of the magnet.

Part (a)

Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on the axis.

Part (b)

Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on the equatorial lines (normal bisector) of the magnet.

Conclusion

Magnetic moment of the bar magnet, $M = 0.48 \text{ J T}^{-1}$

Distance, $d = 10 \text{ cm} = 0.1 \text{ m}$

The magnetic field at distance d , from the centre of the magnet on the axis is given by the relation:

$$B = \frac{\mu_0}{4\pi} \frac{2M}{d^3}$$

Where,

$$\mu_0 = \text{Permeability of free space} = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$$

$$\begin{aligned}\therefore B &= \frac{4\pi \times 10^{-7} \times 2 \times 0.48}{4\pi \times (0.1)^3} \\ &= 0.96 \times 10^{-4} \text{ T} = 0.96 \text{ G}\end{aligned}$$

The magnetic field is along the S-N direction.

The magnetic field at a distance of 10 cm (i.e., $d = 0.1 \text{ m}$) on the equatorial line of the magnet is given as:

$$\begin{aligned}B &= \frac{\mu_0 \times M}{4\pi \times d^3} \\ &= \frac{4\pi \times 10^{-7} \times 0.48}{4\pi(0.1)^3} \\ &= 0.48 \text{ G}\end{aligned}$$

The magnetic field is along the N-S direction.

axis: 0.96 G, S-N direction; equatorial: 0.48 G, N-S direction //

Q

1

Three quantities specifying earth's field



Answer the following questions regarding earth's magnetism: A vector needs three quantities for its specification. Name the three independent quantities conventionally used to specify the earth's magnetic field. The angle of dip at a location in southern India is about 18° . Would you expect a greater or smaller dip angle in Britain? If you made a map of magnetic field lines at Melbourne in Australia, would the lines seem to go into the ground or come out of the ground? In which direction would a compass free to move in the vertical plane point to, if located right on the geomagnetic north or south pole? The earth's field, it is claimed, roughly approximates the field due to a dipole of magnetic moment $8 \times 10^{22} \text{ J T}^{-1}$ located at its centre. Check the order of magnitude of this number in some way. (f) Geologists claim that besides the main magnetic N-S poles, there are several local poles on the earth's surface oriented in different directions. How is such a thing possible at all?

Conclusion

The three independent quantities conventionally used for specifying earth's magnetic field are:

Magnetic declination,

Angle of dip, and

Horizontal component of earth's magnetic field

(b) The angle of dip at a point depends on how far the point is located with respect to the North Pole or the South Pole.

The angle of dip would be greater in Britain (it is about 70°) than in southern India because the location of Britain on the globe is closer to the magnetic North Pole.

(c) It is hypothetically considered that a huge bar magnet is dipped inside earth with its north pole near the geographic South Pole and its south pole near the geographic North Pole.

Magnetic field lines emanate from a magnetic north pole and terminate at a magnetic south pole.

Hence, in a map depicting earth's magnetic field lines, the field lines at Melbourne, Australia would seem to come out of the ground.

(d) If a compass is located on the geomagnetic North Pole or South Pole, then the compass will be free to move in the horizontal plane while earth's field is exactly vertical to the magnetic poles.

In such a case, the compass can point in any direction.

(e) Magnetic moment, $M = 8 \times 10^{22} \text{ J T}^{-1}$

Radius of earth, $r = 6.4 \times 10^6 \text{ m}$

Magnetic field strength, $B = \frac{\mu_0 M}{4 \pi r^3}$

Where,

μ_0 = Permeability of free space $= 4 \pi \times 10^{-7} \text{ Tm A}^{-1}$

$\therefore B = \frac{4 \pi \times 10^{-7} \times 8 \times 10^{22}}{4 \pi \times (6.4 \times 10^6)^3} = 0.3 \text{ G}$

This quantity is of the order of magnitude of the observed field on earth.

(f) Yes, there are several local poles on earth's surface oriented in different directions.

A magnetised mineral deposit is an example of a local N-S pole.

Q

2

Conceptual
questions on
earth's
magnetism



Answer the following questions: The earth's magnetic field varies from point to point in space. Does it also change with time? If so, on what time scale does it change appreciably? The earth's core is known to contain iron. Yet geologists do not regard this as a source of the earth's magnetism. Why? The charged currents in the outer conducting regions of the earth's core are thought to be responsible for earth's magnetism. What might be the 'battery' (i.e., the source of energy) to sustain these currents? The earth may have even reversed the direction of its field several times during its history of 4 to 5 billion years. How can geologists know about the earth's field in such distant past? The earth's field departs from its dipole shape substantially at large distances (greater than about 30,000 km). What agencies may be responsible for this distortion? (f) Interstellar space has an extremely weak magnetic field of the order of 10-12 T. Can such a weak field be of any significant consequence? Explain. [Note: Exercise 5.2 is meant mainly to arouse your curiosity. Answers to some questions above are tentative or unknown. Brief answers wherever possible are given at the end. For details, you should consult a good text on geomagnetism.]

Conclusion

Earth's magnetic field changes with time.

It takes a few hundred years to change by an appreciable amount.

The variation in earth's magnetic field with the time cannot be neglected.

(b) Earth's core contains molten iron.

This form of iron is not ferromagnetic.

Hence, this is not considered as a source of earth's magnetism.

(c) The radioactivity in earth's interior is the source of energy that sustains the currents in the outer conducting regions of earth's core.

These charged currents are considered to be responsible for earth's magnetism.

(d) Earth reversed the direction of its field several times during its history of 4 to 5 billion years.

These magnetic fields got weakly recorded in rocks during their solidification.

One can get clues about the geomagnetic history from the analysis of this rock magnetism.

(e) Earth's field departs from its dipole shape substantially at large distances (greater than about 30,000 km) because of the presence of the ionosphere.

In this region, earth's field gets modified because of the field of single ions.

While in motion, these ions produce the magnetic field associated with them.

(f) An extremely weak magnetic field can bend charged particles moving in a circle.

This may not be noticeable for a large radius path.

With reference to the gigantic interstellar space, the deflection can affect the passage of charged particles.

Q

9

Moment of inertia of an oscillating coil

()

A circular coil of 16 turns and radius 10 cm carrying a current of 0.75 A rests with its plane normal to an external field of magnitude 5.0×10^{-2} T. The coil is free to turn about an axis in its plane perpendicular to the field direction. When the coil is turned slightly and released, it oscillates about its stable equilibrium with a frequency of 2.0 s^{-1} . What is the moment of inertia of the coil about its axis of rotation?

Given

Number of turns in the circular coil, $N = 16$ Radius of the coil, $r = 10 \text{ cm} = 0.1 \text{ m}$ Cross-section of the coil, $A = \pi r^2 = \pi \times (0.1)^2 \text{ m}^2$ Current in the coil, $I = 0.75 \text{ A}$ Magnetic field strength, $B = 5.0 \times 10^{-2} \text{ T}$ Frequency of oscillations of the coil, $\nu = 2.0 \text{ s}^{-1}$

Substitute

$$\begin{aligned} \therefore \text{Magnetic moment, } M &= N I A = N I \pi r^2 \\ &= 16 \times 0.75 \times \pi \times (0.1)^2 \\ &= 0.377 \text{ J T}^{-1} \end{aligned}$$

Key formula

Frequency is given by the relation:

$$\nu = \frac{1}{2\pi} \sqrt{\frac{MB}{I}}$$

Where,

 I = Moment of inertia of the coil

$$\begin{aligned} \therefore I &= \frac{MB}{4\pi^2 \nu^2} \\ &= \frac{0.377 \times 5 \times 10^{-2}}{4\pi^2 \times (2)^2} \\ &= 1.19 \times 10^{-4} \text{ kg m}^2 \end{aligned}$$

Conclusion

Hence, the moment of inertia of the coil about its axis of rotation is $1.19 \times 10^{-4} \text{ kg m}^2$.

$$\boxed{1.19 \times 10^{-4} \text{ kg m}^2} //$$

Q

10

Earth's field from dip angle and horizontal component



A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip pointing down at 22° with the horizontal. The horizontal component of the earth's magnetic field at the place is known to be 0.35 G. Determine the magnitude of the earth's magnetic field at the place.

Given

Horizontal component of earth's magnetic field, $B_H = 0.35 \text{ G}$ Angle made by the needle with the horizontal plane = Angle of dip $= \delta = 22^\circ$ Earth's magnetic field strength = B

Key formula

We can relate B and B_H as:

$$B_H = B \cos \theta$$

$$\therefore B = \frac{B_H}{\cos \delta}$$

$$= \frac{0.35}{\cos 22^\circ} = 0.377 \text{ G}$$

Conclusion

Hence, the strength of earth's magnetic field at the given location is 0.377 G.

0.377 G //

Q

11

Earth's field from declination and dip



At a certain location in Africa, a compass points 12° west of the geographic north. The north tip of the magnetic needle of a dip circle placed in the plane of magnetic meridian points 60° above the horizontal. The horizontal component of the earth's field is measured to be 0.16 G. Specify the direction and magnitude of the earth's field at the location.

Given

Angle of declination, $\theta = 12^\circ$

Angle of dip, $\delta = 60^\circ$

Horizontal component of earth's magnetic field, $B_H = 0.16 \text{ G}$

Earth's magnetic field at the given location = B

Key formula

We can relate B and B_H as:

$$B_H = B \cos \delta$$

$$\therefore B = \frac{B_H}{\cos \delta}$$

$$= \frac{0.16}{\cos 60^\circ} = 0.32 \text{ G}$$

Conclusion

Earth's magnetic field lies in the vertical plane, 12° West of the geographic meridian, making an angle of 60° (upward) with the horizontal direction.

Its magnitude is 0.32 G.

0.32 G, in the vertical plane 12° West of geographic meridian, 60° above horizontal

//

Q

13

Total field at a magnet's null-point distance



A short bar magnet placed in a horizontal plane has its axis aligned along the magnetic north-south direction. Null points are found on the axis of the magnet at 14 cm from the centre of the magnet. The earth's magnetic field at the place is 0.36 G and the angle of dip is zero. What is the total magnetic field on the normal bisector of the magnet at the same distance as the null-point (i.e., 14 cm) from the centre of the magnet? (At null points, field due to a magnet is equal and opposite to the horizontal component of earth's magnetic field.)

Conclusion

Earth's magnetic field at the given place, $H = 0.36 \text{ G}$

The magnetic field at a distance d, on the axis of the magnet is given as:

$$B_1 = \frac{\mu_0}{4\pi} \frac{2M}{d^3} = H$$

Where,

μ_0 = Permeability of free space

M = Magnetic moment

The magnetic field at the same distance d, on the equatorial line of the magnet is given as:

$$B_2 = \frac{\mu_0 M}{4\pi d^3} = \frac{H}{2} \quad [\text{Using equation (i)}]$$

Total magnetic field, $B = B_1 + B_2$

$$\begin{aligned} &= H + \frac{H}{2} \\ &= 0.36 + 0.18 = 0.54 \text{ G} \end{aligned}$$

Hence, the magnetic field is 0.54 G in the direction of earth's magnetic field.

0.54 G, in the direction of earth's magnetic field //

Q

14

New null points after reversing a magnet

(())

If the bar magnet in exercise 5.13 is turned around by 180°, where will the new null points be located?

Conclusion

The magnetic field on the axis of the magnet at a distance $d_1 = 14$ cm, can be written as:

$$B_1 = \frac{\mu_0 2M}{4\pi(d_1)^3} = H$$

Where,

M = Magnetic moment

μ_0 = Permeability of free space

H = Horizontal component of the magnetic field at d_1

If the bar magnet is turned through 180°, then the neutral point will lie on the equatorial line.

Hence, the magnetic field at a distance d_2 , on the equatorial line of the magnet can be written as:

$$B_2 = \frac{\mu_0 M}{4\pi(d_2)^3} = H$$

Equating equations (1) and (2), we get:

$$\begin{aligned} \frac{2}{(d_1)^3} &= \frac{1}{(d_2)^3} \\ \left(\frac{d_2}{d_1}\right)^3 &= \frac{1}{2} \\ \therefore d_2 &= d_1 \times \left(\frac{1}{2}\right)^{\frac{1}{3}} \\ &= 14 \times 0.794 = 11.1 \text{ cm} \end{aligned}$$

The new null points will be located 11.1 cm on the normal bisector.

Q

15

Distance for
a 45-degree
resultant
field



A short bar magnet of magnetic moment $5.25 \times 10^{-2} \text{ J T}^{-1}$ is placed with its axis perpendicular to the earth's field direction. At what distance from the centre of the magnet, the resultant field is inclined at 45° with earth's field on its normal bisector and (b) its axis. Magnitude of the earth's field at the place is given to be 0.42 G. Ignore the length of the magnet in comparison to the distances involved.

Conclusion

Magnetic moment of the bar magnet, $M = 5.25 \times 10^{-2} \text{ J T}^{-1}$

Magnitude of earth's magnetic field at a place, $H = 0.42 \text{ G} = 0.42 \times 10^{-4} \text{ T}$

The magnetic field at a distance R from the centre of the magnet on the normal bisector is given by the relation:

$$B = \frac{\mu_0 M}{4\pi R^3}$$

Where,

μ_0 = Permeability of free space $= 4\pi \times 10^{-7} \text{ Tm A}^{-1}$

When the resultant field is inclined at 45° with earth's field, $B = H$

$$\therefore \frac{\mu_0 M}{4\pi R^3} = H = 0.42 \times 10^{-4}$$

$$\begin{aligned} R^3 &= \frac{\mu_0 M}{0.42 \times 10^{-4} \times 4\pi} \\ &= \frac{4\pi \times 10^{-7} \times 5.25 \times 10^{-2}}{4\pi \times 0.42 \times 10^{-4}} = 12.5 \times 10^{-5} \\ \therefore R &= 0.05 \text{ m} = 5 \text{ cm} \end{aligned}$$

The magnetic field at a distance R' from the centre of the magnet on its axis is given as:

$$B' = \frac{\mu_0 2M}{4\pi R'^3}$$

The resultant field is inclined at 45° with earth's field.

$$\therefore B' = H$$

$$\frac{\mu_0 2M}{4\pi (R')^3} = H$$

$$\begin{aligned} (R')^3 &= \frac{\mu_0 2M}{4\pi \times H} \\ &= \frac{4\pi \times 10^{-7} \times 2 \times 5.25 \times 10^{-2}}{4\pi \times 0.42 \times 10^{-4}} = 25 \times 10^{-5} \\ \therefore R' &= 0.063 \text{ m} = 6.3 \text{ cm} \end{aligned}$$

normal bisector: $R = 5 \text{ cm}$; axis: $R' = 6.3 \text{ cm}$ //

Q

16

Conceptual
questions on
dia-, para-
and
ferromagnetism



Answer the following questions: Why does a paramagnetic sample display greater magnetisation (for the same magnetising field) when cooled? Why is diamagnetism, in contrast, almost independent of temperature? If a toroid uses bismuth for its core, will the field in the core be (slightly) greater or (slightly) less than when the core is empty? Is the permeability of a ferromagnetic material independent of the magnetic field? If not, is it more for lower or higher fields? Magnetic field lines are always nearly normal to the surface of a ferromagnet at every point. (This fact is analogous to the static electric field lines being normal to the surface of a conductor at every point.) Why? (f) Would the maximum possible magnetisation of a paramagnetic sample be of the same order of magnitude as the magnetization of a ferromagnet?

Conclusion

(a) Random thermal motion disrupts the alignment of dipoles at high temperature.

Cooling reduces this disruption, so a paramagnetic sample shows greater magnetisation when cooled.

(b) In a diamagnetic substance, the induced dipole moment always opposes the magnetising field.

So the atoms' internal motion, which depends on temperature, does not affect diamagnetism.

(c) Bismuth is diamagnetic.

So a toroid with a bismuth core has a magnetic field slightly greater than a toroid with an empty core.

(d) The permeability of a ferromagnetic material depends on the applied field: it is greater for a lower field, and vice versa.

(e) A ferromagnetic material's permeability is always greater than one.

That is why magnetic field lines stay nearly normal to its surface at every point.

(f) A paramagnetic sample's maximum possible magnetisation can be of the same order as a ferromagnet's, but this needs a very high magnetising field to reach saturation.

Q

17

Hysteresis,
memory
storage and
magnetic
shielding

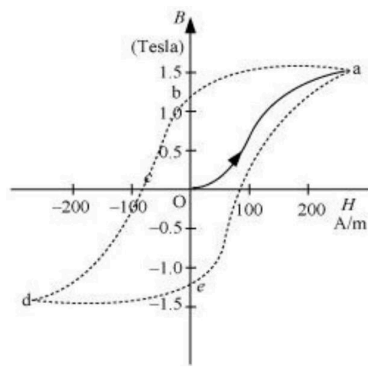


Answer the following questions: Explain qualitatively on the basis of domain picture the irreversibility in the magnetisation curve of a ferromagnet. The hysteresis loop of a soft iron piece has a much smaller area than that of a carbon steel piece. If the material is to go through repeated cycles of magnetisation, which piece will dissipate greater heat energy? 'A system displaying a hysteresis loop such as a ferromagnet, is a device for storing memory?' Explain the meaning of this statement. What kind of ferromagnetic material is used for coating magnetic tapes in a cassette player, or for building 'memory stores' in a modern computer? A certain region of space is to be shielded from magnetic fields. Suggest a method.

Given

The hysteresis curve (B-H curve) of a ferromagnetic material is shown in the figure below.





Conclusion

The curve shows that magnetisation persists even after the external field is removed - this is the irreversibility of a ferromagnet.

(b) The heat dissipated is directly proportional to the area of the hysteresis loop.

A carbon steel piece has a greater loop area, so it dissipates more heat.

(c) The magnetisation value is a record of the hysteresis cycle it went through - these are bits of information tied to that cycle.

So a hysteresis loop can be used to store information.

(d) Ceramic is used to coat magnetic tapes in cassette players and to build memory stores in modern computers.

(e) A region of space can be shielded from magnetic fields by surrounding it with soft iron rings, which draw the magnetic field lines out of that region.

Q

18

Neutral points near a current-carrying cable

((

A long straight horizontal cable carries a current of 2.5 A in the direction 10° south of west to 10° north of east. The magnetic meridian of the place happens to be 10° west of the geographic meridian. The earth's magnetic field at the location is 0.33 G, and the angle of dip is zero. Locate the line of neutral points (ignore the thickness of the cable). (At neutral points, magnetic field due to a current-carrying cable is equal and opposite to the horizontal component of earth's magnetic field.)

Conclusion

Current in the wire, $I = 2.5 \text{ A}$

Angle of dip at the given location on earth, $\delta = 0^\circ$

Earth's magnetic field, $H = 0.33 \text{ G} = 0.33 \times 10^{-4} \text{ T}$

The horizontal component of earth's magnetic field is given as:

$$\begin{aligned} H_H &= H \cos \delta \\ &= 0.33 \times 10^{-4} \times \cos 0^\circ = 0.33 \times 10^{-4} \text{ T} \end{aligned}$$

The magnetic field at the neutral point at a distance R from the cable is given by the relation:

$$H_H = \frac{\mu_0 I}{2 \pi R}$$

Where,

$$\mu_0 = \text{Permeability of free space} = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$$

$$\begin{aligned}\therefore R &= \frac{\mu_0 I}{2\pi H_H} \\ &= \frac{4\pi \times 10^{-7} \times 2.5}{2\pi \times 0.33 \times 10^{-4}} = 15.15 \times 10^{-3} \text{ m} = 1.51 \text{ cm}\end{aligned}$$

Hence, a set of neutral points parallel to and above the cable are located at a normal distance of 1.51 cm.

1.51 cm, parallel to and above the cable //

Q

19

Resultant field near a telephone cable



A telephone cable at a place has four long straight horizontal wires carrying a current of 1.0 A in the same direction east to west. The earth's magnetic field at the place is 0.39 G, and the angle of dip is 35° . The magnetic declination is nearly zero. What are the resultant magnetic fields at points 4.0 cm below the cable?

Conclusion

Number of horizontal wires in the telephone cable, $n = 4$

Current in each wire, $I = 1.0 \text{ A}$

Earth's magnetic field at a location, $H = 0.39 \text{ G} = 0.39 \times 10^{-4} \text{ T}$

Angle of dip at the location, $\delta = 35^\circ$

Angle of declination, $\theta \sim 0^\circ$

For a point 4 cm below the cable:

Distance, $r = 4 \text{ cm} = 0.04 \text{ m}$

The horizontal component of earth's magnetic field can be written as:

$$H_h = H \cos \delta$$

Where,

B = Magnetic field at 4 cm due to current I in the four wires

$$= 4 \times \frac{\mu_0 I}{2\pi r}$$

$$\mu_0 = \text{Permeability of free space} = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$$

$$\begin{aligned}\therefore B &= 4 \times \frac{4\pi \times 10^{-7} \times 1}{2\pi \times 0.04} \\ &= 0.2 \times 10^{-4} \text{ T} = 0.2 \text{ G} \\ \therefore H_h &= 0.39 \cos 35^\circ - 0.2 \\ &= 0.39 \times 0.819 - 0.2 \approx 0.12 \text{ G}\end{aligned}$$

The vertical component of earth's magnetic field is given as:

$$\begin{aligned}H_v &= H \sin \delta \\ &= 0.39 \sin 35^\circ = 0.22 \text{ G}\end{aligned}$$

The angle made by the field with its horizontal component is given as:

$$\begin{aligned}\theta &= \tan^{-1} \frac{H_v}{H_h} \\ &= \tan^{-1} \frac{0.22}{0.12} = 61.39^\circ\end{aligned}$$

The resultant field at the point is given as:

$$H_1 = \sqrt{(H_v)^2 + (H_h)^2}$$

$$= \sqrt{(0.22)^2 + (0.12)^2} = 0.25 \text{ G}$$

For a point 4 cm above the cable:

Horizontal component of earth's magnetic field:

$$H_h = H \cos \delta$$

$$= 0.39 \cos 35^\circ = 0.32 \text{ G}$$

Vertical component of earth's magnetic field:

$$H_v = H \sin \delta$$

$$= 0.39 \sin 35^\circ = 0.22 \text{ G}$$

Angle, θ

$$= \tan^{-1} \frac{H_v}{H_h} = \tan^{-1} \frac{0.22}{0.32} = 34.7^\circ$$

And resultant field:

$$H_2 = \sqrt{(H_v)^2 + (H_h)^2}$$

$$= \sqrt{(0.22)^2 + (0.32)^2} = 0.39 \text{ G}$$

4 cm below: 0.25 G; 4 cm above: 0.39 G //

Q

20

Horizontal
field from a
compass-
deflecting
coil



A compass needle free to turn in a horizontal plane is placed at the centre of circular coil of 30 turns and radius 12 cm. The coil is in a vertical plane making an angle of 45° with the magnetic meridian. When the current in the coil is 0.35 A, the needle points west to east. Determine the horizontal component of the earth's magnetic field at the location. The current in the coil is reversed, and the coil is rotated about its vertical axis by an angle of 90° in the anticlockwise sense looking from above. Predict the direction of the needle. Take the magnetic declination at the places to be zero.

Conclusion

Number of turns in the circular coil, $N = 30$

Radius of the circular coil, $r = 12 \text{ cm} = 0.12 \text{ m}$

Current in the coil, $I = 0.35 \text{ A}$

Angle of dip, $\delta = 45^\circ$

The magnetic field due to current I , at a distance r , is given as:

$$B = \frac{\mu_0 2 \pi N I}{4 \pi r}$$

Where,

μ_0 = Permeability of free space $= 4 \pi \times 10^{-7} \text{ Tm A}^{-1}$

$$\therefore B = \frac{4 \pi \times 10^{-7} \times 2 \pi \times 30 \times 0.35}{4 \pi \times 0.12}$$

$$= 5.49 \times 10^{-5} \text{ T}$$

The compass needle points from West to East.

Hence, the horizontal component of earth's magnetic field is given as:

$$B_H = B \sin \delta$$

$$= 5.49 \times 10^{-5} \sin 45^\circ = 3.88 \times 10^{-5} \text{ T} = 0.388 \text{ G}$$

When the current in the coil is reversed and the coil is rotated about its vertical axis by an angle of 90° , the needle will reverse its original direction.

In this case, the needle will point from East to West.

$B_H = 0.388 \text{ G}$; needle points East to West after reversal //

Q

21

Second field magnitude for a dipole's equilibrium



A magnetic dipole is under the influence of two magnetic fields. The angle between the field directions is 60° , and one of the fields has a magnitude of $1.2 \times 10^{-2} \text{ T}$. If the dipole comes to stable equilibrium at an angle of 15° with this field, what is the magnitude of the other field?

Given

Magnitude of one of the magnetic fields, $B_1 = 1.2 \times 10^{-2} \text{ T}$

Magnitude of the other magnetic field = B_2

Angle between the two fields, $\theta = 60^\circ$

At stable equilibrium, the angle between the dipole and field B_1 , $\theta_1 = 15^\circ$

Angle between the dipole and field B_2 , $\theta_2 = \theta - \theta_1 = 60^\circ - 15^\circ = 45^\circ$

Key formula

At rotational equilibrium, the torques between both the fields must balance each other.

$\therefore$ Torque due to field B_1 = Torque due to field B_2

$M B_1 \sin \theta_1 = M B_2 \sin \theta_2$

Where,

M = Magnetic moment of the dipole

$$\therefore B_2 = \frac{B_1 \sin \theta_1}{\sin \theta_2}$$

$$= \frac{1.2 \times 10^{-2} \times \sin 15^\circ}{\sin 45^\circ} = 4.39 \times 10^{-3} \text{ T}$$

Conclusion

Hence, the magnitude of the other magnetic field is $4.39 \times 10^{-3} \text{ T}$.

$4.39 \times 10^{-3} \text{ T}$ //

Q

22

Deflection of an electron beam in earth's field

((

A monoenergetic (18 keV) electron beam initially in the horizontal direction is subjected to a horizontal magnetic field of 0.04 G normal to the initial direction. Estimate the up or down deflection of the beam over a distance of 30 cm ($m_e = 9.11 \times 10^{-31} \text{ kg}$). [Note: Data in this exercise are so chosen that the answer will give you an idea of the effect of earth's magnetic field on the motion of the electron beam from the electron gun to the screen in a TV set.]

Conclusion

Energy of an electron beam, $E = 18 \text{ keV} = 18 \times 10^3 \text{ eV}$

Charge on an electron, $e = 1.6 \times 10^{-19} \text{ C}$

$$E = 18 \times 10^3 \times 1.6 \times 10^{-19} \text{ J}$$

Magnetic field, $B = 0.04 \text{ G}$

Mass of an electron, $m_e = 9.11 \times 10^{-31} \text{ kg}$

Distance up to which the electron beam travels, $d = 30 \text{ cm} = 0.3 \text{ m}$

We can write the kinetic energy of the electron beam as:

$$E = \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{2E}{m}}$$

$$= \sqrt{\frac{2 \times 18 \times 10^3 \times 1.6 \times 10^{-19} \times 10^{-15}}{9.11 \times 10^{-31}}} = 0.795 \times 10^8 \text{ m/s}$$

The electron beam deflects along a circular path of radius, r .

The force due to the magnetic field balances the centripetal force of the path.

$$B e v = \frac{m v^2}{r}$$

$$\therefore r = \frac{m v}{B e}$$

$$= \frac{9.11 \times 10^{-31} \times 0.795 \times 10^8}{0.4 \times 10^{-4} \times 1.6 \times 10^{-19}} = 11.3 \text{ m}$$

Let the up and down deflection of the electron beam be $x = r(1 - \cos \theta)$

Where,

θ = Angle of declination

$$\sin \theta = \frac{d}{r}$$

$$= \frac{0.3}{11.3}$$

$$\theta = \sin^{-1} \frac{0.3}{11.3} = 1.521^\circ$$

And $x = 11.3(1 - \cos 1.521^\circ)$

$$= 0.0039 \text{ m} = 3.9 \text{ mm}$$

Therefore, the up and down deflection of the beam is 3.9 mm.

3.9 mm //

Q

23

Dipole
moment via
Curie's law



A sample of paramagnetic salt contains 2.0×10^{24} atomic dipoles each of dipole moment $1.5 \times 10^{-23} \text{ J T}^{-1}$. The sample is placed under a homogeneous magnetic field of 0.64 T , and cooled to a temperature of 4.2 K . The degree of magnetic saturation achieved is equal to 15% . What is the total dipole moment of the sample for a magnetic field of 0.98 T and a temperature of 2.8 K ? (Assume Curie's law)

Given

Number of atomic dipoles, $n = 2.0 \times 10^{24}$

Dipole moment of each atomic dipole, $M = 1.5 \times 10^{-23} \text{ J T}^{-1}$

When the magnetic field, $B_1 = 0.64 \text{ T}$

The sample is cooled to a temperature, $T_1 = 4.2^\circ \text{ K}$

Substitute

Total dipole moment of the atomic dipole, $M_{\text{tot}} = n \times M$

$$\begin{aligned} &= 2 \times 10^{24} \times 1.5 \times 10^{-23} \\ &= 30 \text{ J T}^{-1} \end{aligned}$$

Magnetic saturation is achieved at 15 %.

$$\text{Hence, effective dipole moment, } M_1 = \frac{15}{100} \times 30 = 4.5 \text{ J T}^{-1}$$

When the magnetic field, $B_2 = 0.98 \text{ T}$

Temperature, $T_2 = 2.8^\circ \text{ K}$

Its total dipole moment = M_2

Key formula

According to Curie's law, we have the ratio of two magnetic dipoles as:

$$\begin{aligned} \frac{M_2}{M_1} &= \frac{B_2}{B_1} \times \frac{T_1}{T_2} \\ \therefore M_2 &= \frac{B_2 T_1 M_1}{B_1 T_2} \\ &= \frac{0.98 \times 4.2 \times 4.5}{2.8 \times 0.64} = 10.336 \text{ J T}^{-1} \end{aligned}$$

Conclusion

Therefore, 10.336 J T^{-1} is the total dipole moment of the sample for a magnetic field of 0.98 T and a temperature of 2.8 K .

$$\boxed{10.336 \text{ J T}^{-1}} //$$

Q

24

Magnetic field inside a Rowland ring



A Rowland ring of mean radius 15 cm has 3500 turns of wire wound on a ferromagnetic core of relative permeability 800 . What is the magnetic field B in the core for a magnetising current of 1.2 A ?

Given

Mean radius of a Rowland ring, $r = 15 \text{ cm} = 0.15 \text{ m}$

Number of turns on a ferromagnetic core, $N = 3500$

Relative permeability of the core material, $\mu_r = 800$

Magnetising current, $I = 1.2 \text{ A}$

Key formula

The magnetic field is given by the relation:

$$B = \frac{\mu_r \mu_0 I N}{2 \pi r}$$

Where,

μ_0 = Permeability of free space = $4 \pi \times 10^{-7} \text{ Tm A}^{-1}$

$$B = \frac{800 \times 4 \pi \times 10^{-7} \times 1.2 \times 3500}{2 \pi \times 0.15} = 4.48 \text{ T}$$

Conclusion

Therefore, the magnetic field in the core is 4.48 T .

$$\boxed{4.48 \text{ T}} //$$

Q

25

Classical vs
quantum
spin and
orbital
moments



The magnetic moment vectors μ_s and μ_l associated with the intrinsic spin angular momentum S and orbital angular momentum L , respectively, of an electron are predicted by quantum theory (and verified experimentally to a high accuracy) to be given by:

$$\mu_s = - \frac{e}{m} S,$$

$$\mu_l = - \frac{e}{2m} L$$

Which of these relations is in accordance with the result expected classically? Outline the derivation of the classical result.

Conclusion

The magnetic moment associated with the intrinsic spin angular momentum (S) is given as

The magnetic moment associated with the orbital angular momentum (L) is given as

For current i and area of cross-section A , we have the relation:

Where,

e = Charge of the electron

r = Radius of the circular orbit

T = Time taken to complete one rotation around the circular orbit of radius r

Angular momentum, $L = m v r$

Where,

m = Mass of the electron

v = Velocity of the electron

Dividing equation (1) by equation (2), we get:

Therefore, of the two relations, is in accordance with classical physics.